

Steady-state final configurations, for a grid start (left) and a randomized start (right) :

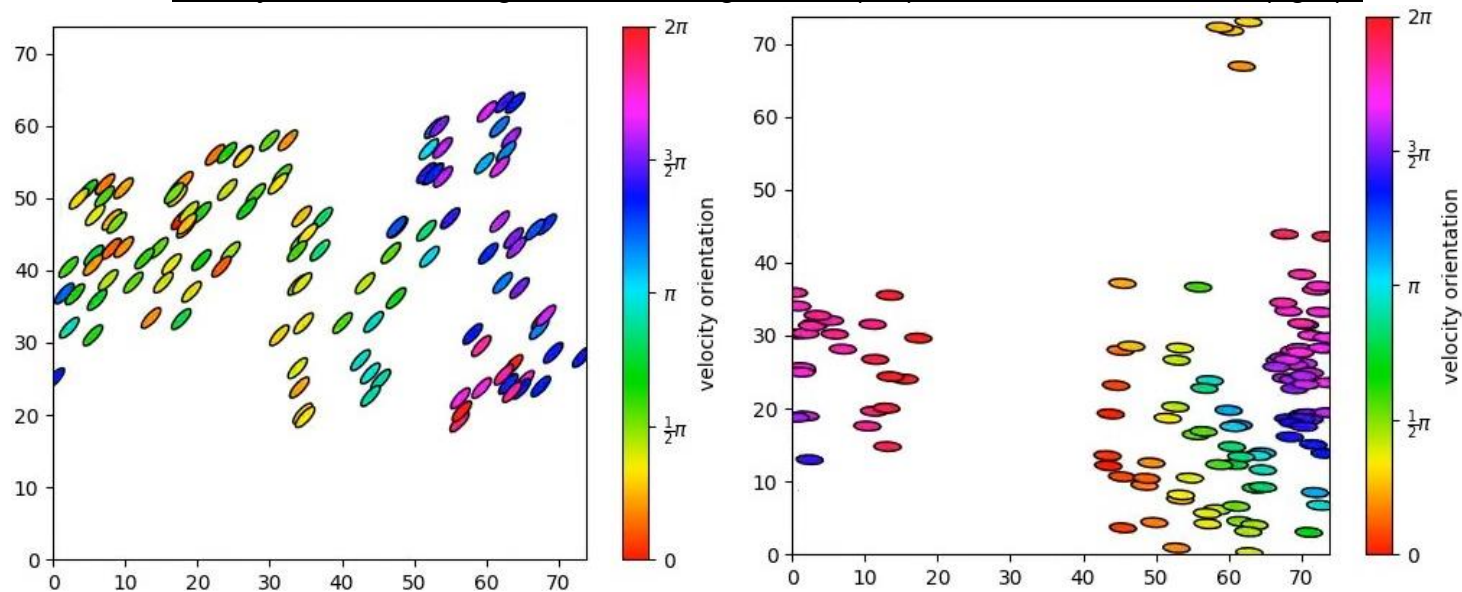


Fig. 1 & 2 : Ordered steady-state configuration (cf code for parameter values)

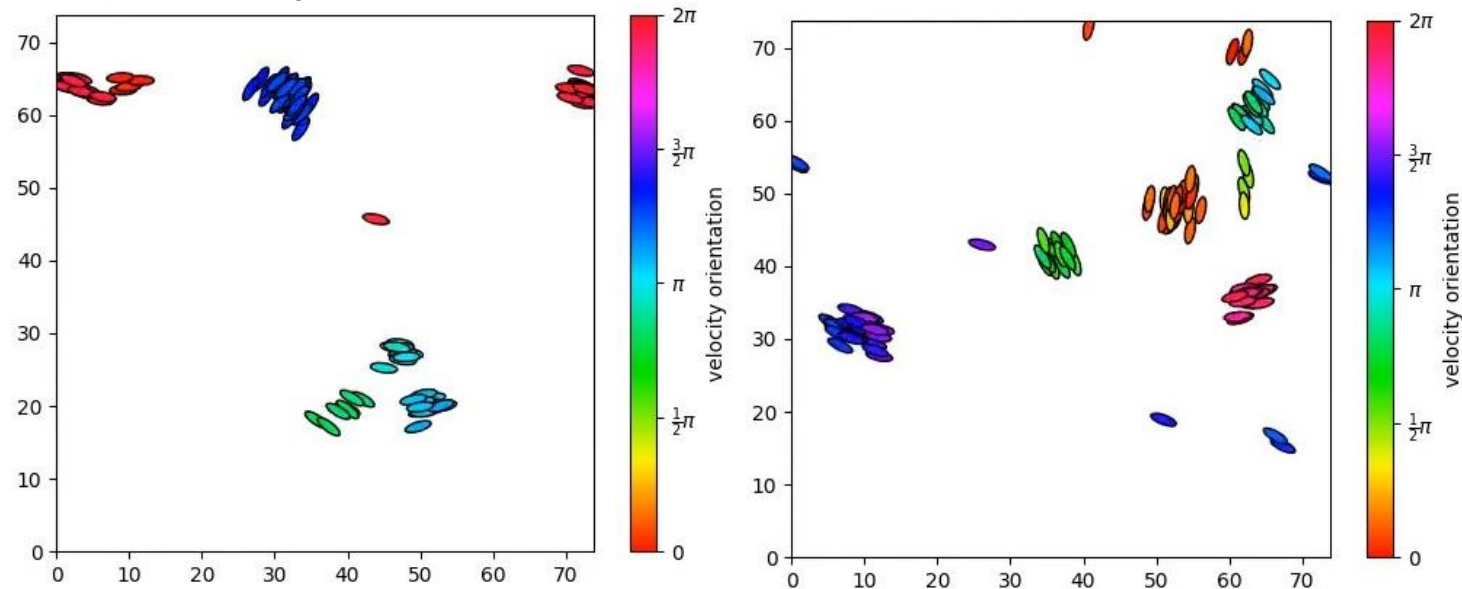


Fig. 3 & 4 : Flocking steady-state configuration (cf code for parameter values)

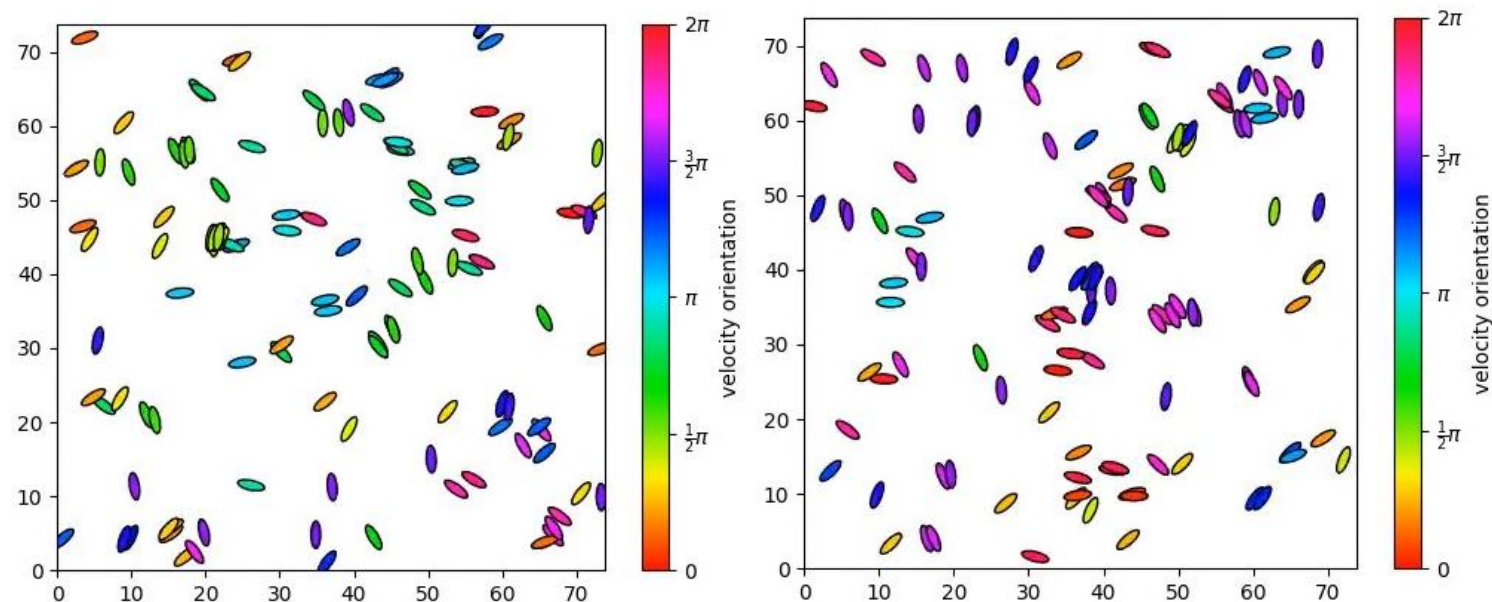


Fig. 5 & 6 : Disordered steady-state configuration (cf code for parameter values)

Average time to update one frame as a function of system size :

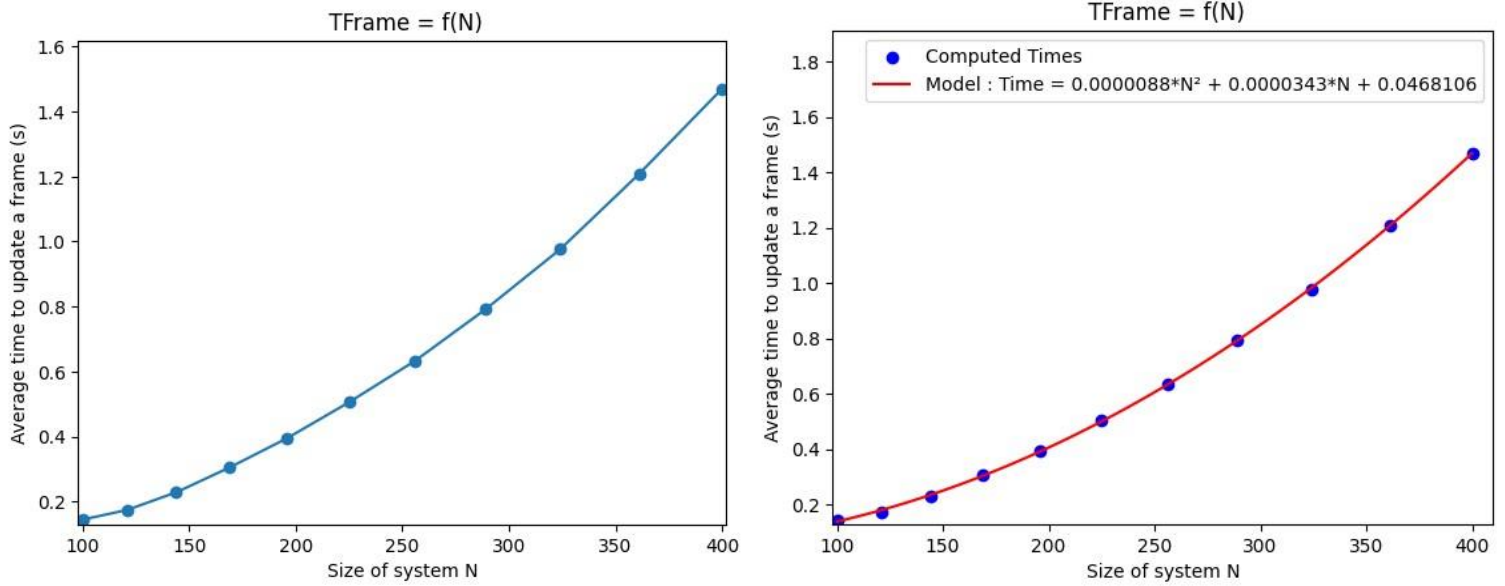


Fig. 7 & 8 : Time to update 1 frame, averaged over 100 frames, with (right) and without (left) polyfit model

Phase transition diagram :

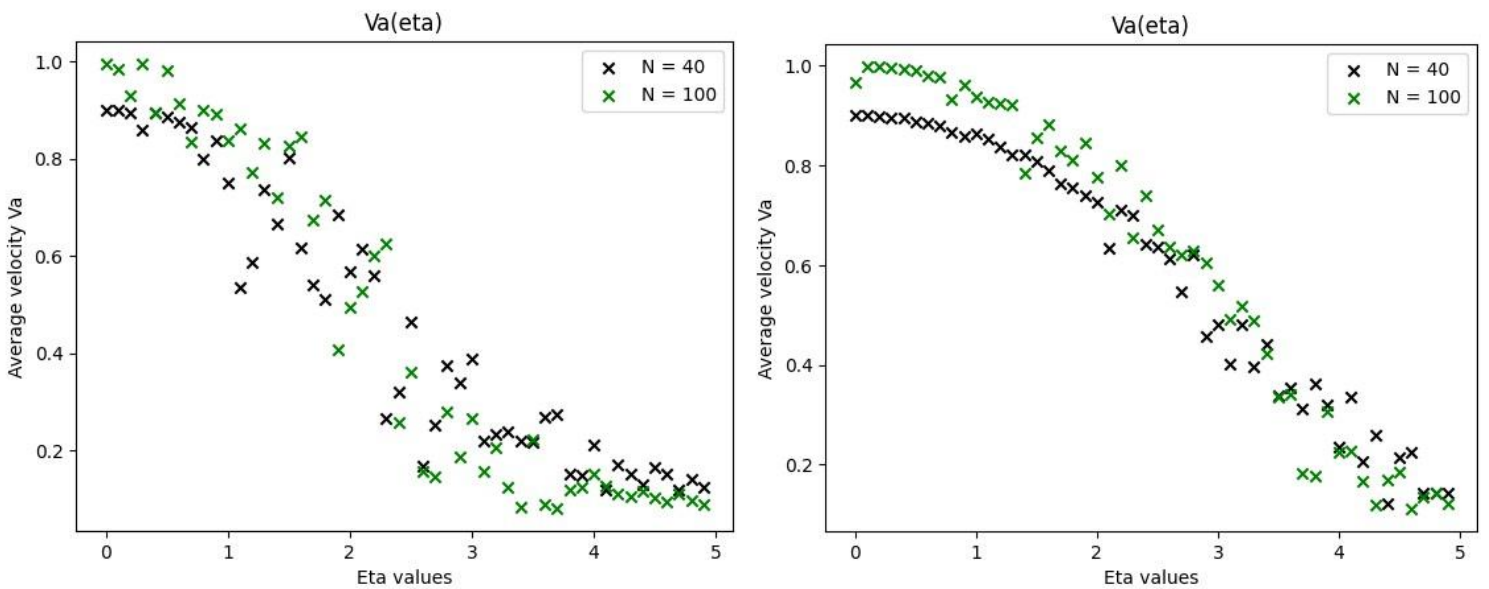


Fig. 9 & 10 : Average normalized velocity in steady-state, averaged over 20 frames, as a function of noise strength η for a vision radius of 1.5 (left) and 3.0 (right)

All of the above graphs and the code used to generate them can be found in this Google Drive folder :

https://drive.google.com/drive/folders/18dDRVc2ND2AbxiMvvy2lqfRJs6sTaD6O?usp=drive_link